# Harvesting Textual and Contrastive Data from the HAL Publication Repository

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## Abstract

Authorship attribution in natural language processing traditionally struggles to distinguish genuine stylistic signals from topical confounds. While contrastive learning approaches have addressed this by maximizing semantic overlap between positive pairs, creating large-scale datasets under strict topic constraints remains challenging. We introduce HALvest, a 17-billion-token multilingual corpus harvested from 778k open-access academic papers, and HALvest-Contrastive, a derived dataset designed to isolate stylometric signals through controlled topic variation. Unlike prior work that minimizes lexical overlap, we exploit natural topic drift between papers by the same author, treating residual lexical patterns as authorial fingerprints rather than noise. Comparing lexical baselines (BM25) against neural models trained on unrestricted (topic-rich) versus base (topic-decoupled) triplets, we demonstrate that models trained exclusively on topic-decoupled data achieve superior performance across all test conditions, outperforming both retrieval baselines and models exposed to topic-rich training data. Our analysis reveals that while lexical signals provide substantial performance gains for keyword-driven methods, neural architectures learn robust stylometric representations that plateau with moderate context length, suggesting they capture distributional style beyond surface-level tokens. Both datasets and code are publicly available.

## 1 Introduction

Authorship attribution, the task of determining whether a given text was written by a specific author, is a long-standing problem in natural language processing (NLP). Verifying authorship has crucial applications in detecting large language model-generated content, plagiarism detection, and the digital humanities. While modern language models have achieved impressive performance on this task (Boukhers and Asundi, 2023), a fundamental question remains: are these models learning an author's unique stylistic fingerprint (stylometry), or do they rely on simpler, more superficial cues?

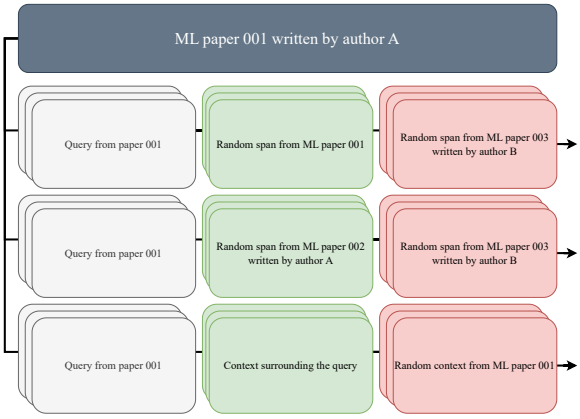


Figure 1: The dataset's different configurations. From top to bottom: full text of an extracted paper from HAL, unrestricted triplets, base triplets, and inverse cloze task (ICT). Positives are in green and negatives in red. In the unrestricted configuration, positive spans can be sampled from the same document as the query.

Repositories for scholarly work like HAL [1] or Google Scholar faces author disambiguation challenges when researchers lack unique identifiers. Papers by authors with common names or homonyms can be misattributed. While disambiguation methods rely primarily on co-author graphs and metadata, textual stylometry could provide complementary signal. This raises a key question: can we extract robust stylistic features that remain consistent across an author's varied research topics?

The primary obstacle is topical confounding. Models can associate documents with authors not by capturing distinct writing style, but by detecting shared, niche terminology from common subject matter. Recently, research in authorship attribution

[1]HAL (https://hal.science/) is a repository of scientific documents, whether published or not, produced in the scientific research and higher education context, especially in France.

has favored contrastive learning approaches (Wegmann et al., 2022; Huertas-Tato et al., 2024; Bevendorff et al., 2021), as they offer more control over experimental conditions and the features to test (Rivera-Soto et al., 2021). The goal was to learn pairs of sentences from the same author and on the same topic, to maximize vocabulary overlap (Wegmann and Nguyen, 2021). Most of this prior work shares the intuition that decreasing the discriminative signal from individual tokens may encourage models to learn more subtle distributional cues (e.g., stop-word frequency, sequence length, or overall tone).

We propose a solution based on the inherent structure of academic papers, where authors write on related but distinct scholarly topics while maintaining consistent stylistic patterns. We construct triplets from HAL documents with two configurations: (1) *unrestricted triplets*, where query and positive can share a document (high topical overlap), and (2) *base (restricted) triplets*, where query and positive come from different papers by the same author (minimal topical overlap). This design directly tests whether models rely on stylistic versus topical signals. For each author, we exploit this intra-author topic variation and introduce controlled confounding by sampling hard negatives from papers within the same field. In this setup, token overlap becomes informative, potentially signaling writing habits rather than shared subject matter.

Our evaluation tests a specific hypothesis: if restricted-triplet training forces models to ignore topical shortcuts, these models should generalize better. We compare models trained on unrestricted data (topic-rich) versus restricted data (topic-poor), plus a lexical baseline (BM25) and an inverse cloze task (ICT) baseline for style-aware retrieval. If the restricted-trained model outperforms, we would have an evidence that the model can rely less on topical cues and more on stable stylistic indicators. We systematically evaluate this hierarchy of models across different test configurations with varying levels of topical confounding.

Our experiments reveal three key findings. First, we show that even a strong lexical baseline (BM25) that performs well on standard, topic-rich data fails when topical cues are removed. Second, we demonstrate that a neural model trained exclusively on restricted triplets learns the most robust representation of style, achieving the highest accuracy across all test conditions. Third, we discuss a critical trade-off between this model's peak performance and the practical challenges of scalability and potential author sampling bias. Motivated by these results, we release a scaled version of the restricted configuration containing millions of triplets [2].

We also introduce HALvest which served as the sampling source for our triplets: 17 billion tokens harvested from HAL's open papers. Although HALvest is mostly in English and French, the gathered papers are written in 56 languages across 16 identified domains. It serves as a valuable asset for low-resource languages, hosting documents in Basque, Catalan, and Persian among others (§B.1). From HALvest, we derive HALvest-Contrastive for contrastive learning at different levels on sentence triplets (Figure 1). We release the original dataset [3] and the code to construct it [4].

## 2 Related Work

**Large-Scale Scholarly Datasets** Realising large-scale textual data from publication repositories has become central to NLP research. Prominent examples include domain-specific corpora from ArXiv, DBLP, and PubMed (Sen et al., 2008; Doğan et al., 2014; Wahle et al., 2022; arXiv.org submitters, 2024). Other initiatives even provide full-text access, such as the ACL Anthology for computational linguistics (Bird et al., 2008), or S2ORC (Lo et al., 2020), notable for its scale and multi-domain nature. Our work introduces HALvest, which contributes to this landscape by providing a fully open, large-scale and multilingual corpus that is also diverse in its document types.

**Contrastive Learning for Stylometry** Traditionally, authorship analysis has been framed as a classification task on sources like online forums (Schler et al., 2006; Seroussi et al., 2014). In fact, existing benchmarks such as the PAN shared tasks (Bevendorff et al., 2021) have been instrumental in driving research, though they are often constrained to a closed set of authors and may have access restrictions. However, more recently, contrastive learning has emerged as a powerful paradigm for stylometry (Wegmann et al., 2022; Huertas-Tato et al., 2024). A prevailing strategy in this area has been to decorrelate an author's style from literal text con-

[2] https://huggingface.co/datasets/almanach/HALvest-Contrastive

[3] https://huggingface.co/datasets/almanach/HALvest

[4] https://github.com/Madjakul/HALvesting

tent. This includes developing edit-invariant loss functions to capture signals beyond surface-level text (Liu et al., 2022), or creating positive pairs with high semantic overlap to force models to learn distributional cues (Rivera-Soto et al., 2021; Wegmann and Nguyen, 2021). In these setups, the goal is often to minimize the usefulness of individual keywords as a signal.

We adopt a different perspective. Instead of minimizing token overlap, we control for topic at a higher level by sampling texts from different papers within the same broad academic field. In our setup, we hypothesize that the remaining lexical overlap is not a confounding variable to be eliminated, but rather a valuable component of an author's stylistic fingerprint.

## 3 Collection Methodology

HALvest construction involves three stages: metadata extraction from HAL, quality filtering, and contrastive triplet generation.

### 3.1 Fetching data from HAL

HAL's API allows for precise queries to the repository, returning complete XML-TEI responses if prompted to. This format heavily facilitate metadata's gathering in a structured format (§ B). We designed our query to only fetch documents flagged as open access with an attached PDF.

We used GROBID (GROBID Repository) to derive structured XML files from the PDFs, easing the conversion to plain text afterward. Reference markers in each document are serialized following the scheme defined by Taylor et al. (2022). However, mathematical annd SMILES formulas, as well as code, are encoded in plain utf-8 and left as is for now.

**Why using GROBID?** Relying on GROBID as a proxy before performing PDF parsing may seem far-fetched, as more accurate ways exist to extract text from PDFs—think of vision-language models and Optical Character Recognition (OCR). However, having a latent structured view of our articles will enable finer control over the desired text output we want in the future. The long-term goal is to be able to standardize and change the format of high-bandwidth text, like citations or code, at will.

### 3.2 Filtering

During the conversion process, if the font within a PDF lacks Unicode tables and do not employ standard encoding for mapping glyph indices to characters, GROBID's output ends up being gibberish (§A.2), as it does not employ OCR to extract text from PDFs, but rather the layout. To filter out gibberish documents, we use a set of heuristic functions:

1. Documents with less than 3 words are discarded as they likely do not provide any valuable information.

2. Documents with more than 10% of words that are capitalized are discarded. Some gibberish documents consist of random capitalized letters. This threshold is low to not filter out documents with a lot of DNA sequences or a lot of tables with capitalized sections.

3. Documents with more than 60% of words that are not alphanumeric are discarded. Some gibberish documents consist of random words with non-alphanumeric characters overrepresented.

4. Documents with an average word length of 1.5 characters or below are discarded. GROBID sometimes escapes the words in a document letter by letter.

5. Documents with no stop words are discarded. Stop words are strong indicators for well-written documents, hence, lowering the chance of it being gibberish. Besides, we use stop words as a language identifier, as the language provided in the metadata is only the one specified by a depositor.

6. We compute an inverse fertility score (Rust et al., 2021): the number of words in a document divided by the number of mt5 tokens (Xue et al., 2021). After removing special tokens, a tokenizer yields at most an amount of tokens equal to the number of words, effectively bounding our function between 0 and 1. An inverse fertility score close to zero is a strong indication of overtokenization, therefore hinting at a gibberish document. We discard documents with an inverse fertility score lower than 0.2.

Following Raffel et al. (2020); Wenzek et al. (2020); Rae et al. (2022) and Penedo et al. (2023), we compute a set of statistics about each document, such as the number of lines, the average word

length or the ratio of unique words. We followed the document quality filtering heuristics from RedPajama Repository project. To compute the ratio of stop-words in a document, we use topwords-json (6, 2024) and found that a near zero ratio of stop-words within a text was a good quality indicator.

We used a coarse filtering process to allow documents with few gibberish sections to remain, though a more fine-grained filtering can leverage a useful signal to pretrain a language model (Antoun et al., 2024).

### 3.3 Contrastive pairs

| # base sentences | Train | Valid | Test |
|---|---|---|---|
| 2 | 1.88M | 19.1k | 19.1k |
| 4 | 1.4M | 14.3k | 14.3k |
| 6 | 1.11M | 11.3k | 11.3k |
| 8 | 892k | 9.1k | 9.1k |
| **# ICT sentences** | | | |
| 1 | 296k | 3.02k | 3.02k |
| 2 | 204k | 2.08k | 2.08k |
| 3 | 174k | 1.78k | 1.78k |
| 4 | 150k | 1.53k | 1.53k |

Table 1: Number of triplets in the HALvest-Contrastive datasets for each configuration. In the base configuration, query, positive and negative spans are sampled independently, with the query and the positive sampled from different documents. In the ICT configuration, the query has $N$ contiguous sentences and the positive is the $2N$ surrounding sentences from the query; the negative is a random context of similar size from the same document.

Given a query, authorship attribution involves retrieving passages from the same author as the query among a set of documents. We mined the said authored passages from HALvest and strongly filtered them. We only considered documents with at least one reference, as a proxy to identify research papers: HAL doesn't only host research papers, but also books or dictionaries for example. This constraint is an indirect control of the tone across the dataset, making it more homogeneous. Finally, we filtered out documents addressing subjects that are not within the 13 identified domains (§B.2) to facilitate subsequent hard negative mining.

- Sequences with too many symbols and math.
- Sequences with abnormal layout, hinting at a figure or a tabular.
- Sequences with code, urls, mail addresses or HTML markups.
- Sequences with a high upper case ratio or header terms like "keywords" or "abstract" as it gives too much context on the topic.
- Sequences with a high variance in sentence lengths, that would indicate abbreviations on initialism flagged as sentence (Fig., or e.g. for example).
- Sequences with zero english stop-words, that would indicate either a gibberish span or a non-english one.

**Unrestricted dataset** Sampling query, positive and negative triplets is a two-step process. For each paper in HALvest, we sample 5 random spans of $N = \{2, 4, 6, 8\}$ sentences as queries. To get the positives, we draw all the documents from the exact same set of authors and randomly sample 5 spans. If there are no other documents by the same authors, the spans are sampled from the same document. This first contrastive dataset allowed us to assess the effect of confounding topic and style before being discarded.

**Base (Restricted) dataset** In this configuration, we only sample positive pairs from authors from different documents. We sample 10 random spans for the queries, as this is our final scaled dataset. In both configurations, for each pair, we mine hard negatives from documents with none of the authors from the query but with the exact same set of HAL domains. Our final corpus is constructed from millions of scholarly documents, written by ≈40,000 unique authors. A key characteristic of this dataset is its inherently collaborative nature, with an average of 1.84 authors per document. While this low collaboration rate and the high document-to-author ratio (≈47) might suggest a clean signal, these top-level statistics can be misleading for our task. A traditional authorship attribution model would be forced to treat co-authorship as significant label noise, as it attempts to map a blended stylistic signal to a single author. We overcome this challenge with contrastive learning: the ground truth label is not an individual author, but rather the *exact, complete set of authors*.

**Inverse cloze task (ICT)** In the ICT setting (Lee et al., 2019; Izacard et al., 2022), we identify a set of $3N$ contiguous sentences $S$. For

$S = \{s_1, s_2, ..., s_a, ..., s_b, ..., s_{3N}\}$, we extract a query $q = \{s_a, ..., s_b\}$ and a passage $p^+ = \{s_1, s_2, ..., s_{a-1}, s_{b+1}, ..., s_{3N}\}$. Following Lee et al. (2019), we leave the query within the context 10% of the time to encourage token matching when possible from models. The hard negatives are non-overlapping passages from the same document. We create 1.63M triplets before applying the same filtering as the base configuration.

## 4 Experiments

### 4.1 Retrieval or stylometry?

To evaluate how well language models embed stylometry compared to relying on topical cues, we conduct a series of experiments on a small unrestricted configuration of HALvest-Contrastive. We establish a hierarchy of models, from a pure lexical baseline to increasingly sophisticated neural architectures, and test them on our unrestricted, base, and ICT triplets. All neural models are based on RoBERTa (Liu et al., 2019) and accuracy is reported as the primary metric.

**Lexical Signal Analysis** We first analyze the inherent lexical signal within our dataset using Jaccard similarity. As shown in Figure 3, the lexical signal is substantially higher in the unrestricted triplets, where the query and positive can come from the same paper. For the 8-sentence split, the mean signal is 0.11 for unrestricted triplets but only 0.022 for the base configuration. This confirms that the base `HALvest-Contrastive` successfully minimizes topical overlap, forcing any successful model to rely primarily on signals other than shared keywords.

**Model Performance** We evaluate four models: a robust BM25 baseline representing pure lexical retrieval; an ICT model trained for style-aware retrieval; a model trained on unrestricted triplets; and a base-restricted model trained exclusively to learn a more topic-invariant representation. The results are presented in Table 2. ICT data trains a model to perform context retrieval, or skip-gram (Mikolov et al., 2013) at the sentence level. The subsequent models are thus more suited for topic-matching but learn a more subtle signal than BM25, making them "style-aware retrieval" systems.

The results in Table 2 reveal a clear hierarchy of performance.

1. **BM25 establishes a strong topic baseline.** It performs reasonably well on the unrestricted set, where topical cues are abundant, but its performance drops significantly on the base set. This confirms that pure lexical retrieval is insufficient when the topical shortcut is removed.

2. **ICT learns style-aware retrieval.** The ICT model consistently outperforms BM25, especially on the base set. This indicates that its retrieval objective learns a more abstract representation that captures stylistic similarities.

3. **The model trained with base data is superior.** Our key finding is that the model trained ith tripelts where we restrict the query and positives to be from different documents, is the top performer across all conditions.

We also evaluate the learned stylometric embeddings on the fully scaled `HALvest-Contrastive` dataset. Given a query document, the goal is to retrieve all other documents from the same author-set from a pool of ≈14,300 candidates. Our results hint at a generalized high-dimensional representation of authorial style, rather than overfitting to high-frequency author-sets. It achieves a normalized discounted cumulative gain (nDCG@10) of 19.84%. While modest in isolation, this score is exceptionally strong given the task's difficulty: a random baseline, which shuffles the 14.3k candidates we had in our test set, would yield an nDCG@10 of $\frac{10}{14300} \approx 0.07\%$. Moreover, comparing retrieval metrics to a model trained with ICT data shows that the model trained with base data significantly outperforms the ICT model, further supporting our hypothesis.

### 4.2 Sequence length

Revisiting the vocabulary analysis in Figure 3, we observe that increasing the sequence length from 2 to 8 sentences amplifies the lexical signal for both triplet types. However, the effect is far more pronounced when triplets are unrestricted. In contrast, the signal for restricted triplets, while also increasing, is substantially weaker and begins to plateau after 6 sentences. While more text provides more stylistic evidence, the amount of unique author-specific evidence in topic-decoupled scenarios seems saturates quickly.

Upon assessing the aforementioned trend (Figure 2 and Table 2), we observe that each model benefits from longer sequences. However, the magnitude of this improvement is revealing. The

| Model | 2 Sentences | | 4 Sentences | | 6 Sentences | | 8 Sentences | |
|---|---|---|---|---|---|---|---|---|
| | Unrestr. | Base | Unrestr. | Base | Unrestr. | Base | Unrestr. | Base |
| BM25 | 68.15 | 60.24 | 79.26 | 68.36 | 85.03 | 73.72 | 89.09 | 78.21 |
| ICT | 86.61 | 79.72 | 91.60 | 85.11 | 93.77 | 88.51 | 94.41 | 88.07 |
| RoBERTa-UnrestrData | – | 80.73 | – | 87.03 | – | 88.44 | – | 89.91 |
| RoBERTa-BaseData | – | **82.33** | – | **88.07** | – | **90.15** | – | **90.83** |

Table 2: Accuracy (%) on the base, unrestricted and ICT test triplets. For a fair comparison, given the smaller initial size of the base configuration, we compare all models after 3.1k training steps.

| | Accuracy | Recall@10 | nDCG@10 |
|---|---|---|---|
| RoBERTa | 65.11 | 3.96 | 3.30 |
| RoBERTa-ICTData | 85.14 | 15.61 | 12.43 |
| RoBERTa-BaseData | 91.59 | 25.47 | 19.84 |

Table 3: Accuracy, Recall@10, and nDCG@10 (%) on the 4-sentence validation set. For a fair comparison given the smaller size of the ICT configuration (after scaling), we compare all models after 5.4k training steps.

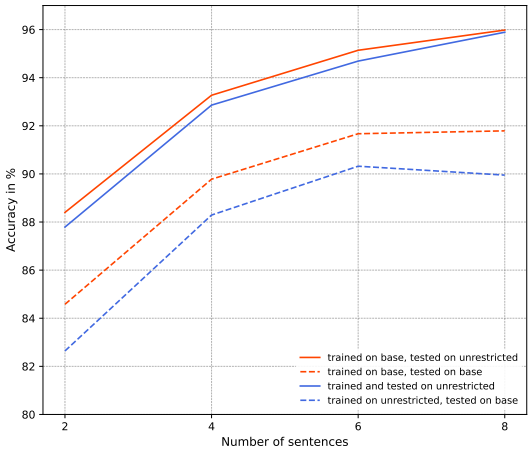


Figure 2: Two fine-tuned language models, in blue and red, respectively trained on unrestricted and base data. Plain lines track performance on unrestricted test data.

purely lexical BM25 model, despite having the lowest overall accuracy, sees the largest relative performance gain (approximately 20% improvement from 2 to 8 sentences in Table 2). BM25 relies heavily on the growing lexical signal; its performance is almost entirely coupled to the availability of more keywords. In contrast, language models exhibit a more modest gain (approximately 10%). The neural models appear to capture a substantial portion of the authorial signal even at shorter sequence lengths, making them less dependent on raw lexical signal alone. While they also leverage the additional lexical cues present in longer texts, we hypothesize that they primarily use the extra context to refine a more complex, non-lexical representation of style. Core stylistic elements (syntax, preferred transition words, clause structures) are captured in the first few sentences, and additional text might provide diminishing returns for style (as opposed to topic).

## 5 Limitations

Our methodology introduces several constraints that warrant discussion.

**PDF processing** GROBID-based processing hindered the dataset creation in some aspects. As only the PDF's layout tokens matter to GROBID, and no optical character recognition is performed, documents with odd layouts cannot be effectively used after conversion, and are outright discarded. Most of the documents are seamlessly converted, yet some of them contain spans of text that are nearly gibberish, embedded within otherwise clean sentences. We designed our filtering process to remove documents with a prior unusable layout, but not to remove the gibberish span of text within clean data. Therefore, further processing effort is needed from practitioners. As a rough estimation, Antoun et al. (2024) preprocessed HALvest's French split at a sentence level, removing tabs and some mathematical formulas, and ended with about 52% of the total tokens (4.7 billion tokens). But this estimation can vary from one language to another, as GROBID does not process Arabic documents correctly, for example, reversing every word.

**Multilingual documents** Some authors write and submit documents in two or more languages, but HAL requires them to only specify one language. Practitioners must perform language identification at the sentence level to ensure only the needed language is fetched from the dataset.

**Co-authorship** A necessary trade-off in our methodology is the ambiguity introduced by defin-

ing labels as author-sets. By training the model to recognize a composite signal, we accept an inherently "noisier" target than a pure, single-author stylometric fingerprint. This approach may limit the model's ability to disentangle and isolate the stylistic features of individual authors. For instance, the embedding for `[Author A, Author B]` represents a unique, negotiated style, which may be distinct from the simple sum of its parts (`[Author A]` + `[Author B]`). This composite signal may make it challenging for a model to generalize to tasks requiring the identification of an individual author's contribution within a novel collaboration (e.g., identifying `Author A`'s presence in an unseen `[Author A, Author C]` document).

**Discriminative keywords** We do not systematically study the relationship between a model performance and the amount of signal and noise in the triplet as it would also depend on the architecture. For instance, going from an absolute positional encoding to a relative one might change the performance and how the model is able to exploit the sentence structure.

## 6 Conclusion

We presented HALvest and HALvest-Contrastive: resources designed to advance research in stylometry by enabling controlled analysis of authorship signals independent of topic. Our experiments establish two key findings. First, lexical baselines perform well when topical cues are abundant but degrade substantially when these shortcuts are removed, confirming that pure keyword matching is insufficient for robust stylometry. Second, neural models trained exclusively on topic-decoupled triplets learn the most robust stylistic representations.

The residual lexical overlap in our approach, rather than being treated as noise, appears to capture meaningful stylistic patterns. Neural models leverage this signal alongside distributional cues, as evidenced by their performance plateau at moderate context lengths, a behavior distinct from purely lexical methods that scale almost linearly with available keywords.

Future work should explore cross-domain generalization of learned stylometric representations, and investigate how well a single author can be distinguished from its co-authors within a document.

## Ethics Statement

In this work, we have taken careful measures to ensure that our dataset is constructed solely from open-access scholarly documents available in the HAL repository. All source materials are either distributed under open licenses or no license at all: despite this variety of licensing, our data collection pipeline strictly adheres to the usage terms provided by HAL, as we do not re-distribute any paper.

Furthermore, our processing workflow—ranging from PDF-to-text conversion with GROBID to subsequent filtering—has been designed to preserve the integrity of the original texts while minimizing the risk of propagating errors. The dataset is intended solely for research purposes, and does not include any sensitive personal information beyond what is publicly accessible.

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# A Collection Methodology

## A.1 API request format

We use the following request to fetch open papers from HAL: https://api.archives-ouvertes.fr/search/?q=*&fq=submittedDate_tdate:[2001-01-01T00:00:00Z%20TO%202024-02-29T23:59:59Z]&fq=openAccess_bool:true&wt=xml-tei&sort=docid%20asc&rows=500&cursorMark=*

## A.2 Example of texts from PDF files with faulty unicode mappings

| halid | Sample Text |
|---|---|
| 01762182 | 1 Introduction 1 IFI wodel nd prolem sttement F F F F F F F F F F F F F F F F F I IFIFI ystem model F F F F F F F F F F F F F F F F F F F F F F I IFIFP vrge dimensionl regime F F F F F F F F F... |
| 00177057 | , 4 @ / A -@ & $ B 4 & A / 2 @ - " 0 / -4 , / 3 $C" ' 5$ B 5C % " +0 2 C % BB * ( ( $ - $ @" ' + 0 D 0 E $ F $&$@" C % < = $ B 7 (#7 $ - $ @" 0 D 0 E @" @ C G $ ( > (( < 5HI 97>89 D (0 E " @C"@F C G $... |
| 01770410 | ACKNOWLEDGMENT S T h i s d o c t o r a l w o r k i s a b o u t m a k i n g o u r s w h a t w e s h a r e . Y e t , I c o u l d n o t h a v e r e a c h e d t h i s f i n a l s t a g e w i t h o u t s h... |
| 01784066 | II Theory L I S T O F T A B L E S Table Deterministic exact approaches to mean-variance portfolio selection problem (see also [START_REF] Mansini \| Twenty Years of Linear Programming Based Portfolio Optimization[END_REF]... |

Table 4: Examples of halid and output text from PDF files with no Unicode mapping, leading to gibberish text. While the first two examples are from discarded documents, the third one is kept, as only the section names are not processed correctly.

# B Composition

We selected eight features to represent the submissions:

- `halid`: submission's unique identifier assigned by HAL.
- `lang`: the language of the document, as filled by the depositor.
- `title` Title of the document.
- `domain`: list of field of studies [5].
- `timestamp`: time of access.
- `year`: publication year of the document if relevant. Otherwise, it is set to year 1.
- `url`: URL to access the PDF.
- `authors`: list of authors.

An author comprises nine features::

- `name`: string for the author name, as filled by the depositor.
- `affiliations`: list of unique identifiers attributed by HAL to the institutions where the author belongs.
- `halauthorid`: unique identifier assigned by HAL to each author registered on the online repository. If an author is not registered to HAL, he is considered unidentified and is assigned a `halauthorid` of "0".

### B.1 Language composition

### B.2 Domain composition

## C Experiments

[5] https://hal.science/browse/domain

| ISO-639 | Language | # Documents | # mT5 Tokens |
|---|---|---|---|
| en | English | 464,679 | 8,158,933,235 |
| fr | French | 199,216 | 9,018,529,985 |
| es | Spanish | 2,975 | 69,221,667 |
| it | Italian | 1,172 | 48,747,986 |
| pt | Portuguese | 934 | 32,918,832 |
| de | German | 652 | 12,225,960 |
| ru | Russian | 245 | 5,763,532 |
| zh | Chinese | 160 | 2,861,585 |
| eu | Basque | 113 | 2,297,485 |
| ar | Arabic | 92 | 2,167,431 |
| ja | Japanese | 92 | 547,861 |
| el | Greek | 54 | 1,738,878 |
| pl | Polish | 43 | 987,878 |
| ro | Romanian | 39 | 1,298,901 |
| uk | Ukrainian | 34 | 837,793 |
| vi | Vietnamese | 29 | 436,660 |
| ca | Catalan | 28 | 975,078 |
| da | Danish | 27 | 961,955 |
| oc | Occitan | 26 | 285,334 |
| br | Breton | 24 | 998,088 |
| sr | Serbian | 24 | 336,878 |
| ko | Korean | 17 | 226,268 |
| fa | Persian | 17 | 213,903 |
| tr | Turkish | 17 | 149,718 |
| hu | Hungarian | 14 | 577,568 |
| eo | Esperanto | 14 | 105,286 |
| hy | Armenian | 10 | 127,988 |
| cs | Czech | 9 | 712,263 |
| bg | Bulgarian | 9 | 208,763 |
| sq | Albanian | 9 | 98,009 |
| id | Indonesian | 9 | 53,075 |
| he | Hebrew | 8 | 61,283 |
| hr | Croatian | 8 | 40,621 |
| et | Estonian | 7 | 20,405 |
| sv | Swedish | 6 | 270,642 |
| no | Norwegian | 6 | 62,767 |
| az | Azerbaijani | 5 | 52,762 |
| fi | Finnish | 4 | 60,507 |
| tet | Tetum | 4 | 18,485 |
| lt | Lithuanian | 3 | 16,572 |
| mr | Marathi | 3 | 16,386 |
| hi | Hindi | 3 | 3,490 |
| ie | Interlingue | 2 | 140,383 |
| ta | Tamil | 2 | 77,087 |
| sw | Swahili | 2 | 73,921 |
| tl | Tagalog | 2 | 35,962 |
| gl | Galician | 2 | 29,688 |
| mk | Macedonian | 2 | 14,654 |
| th | Thai | 1 | 70,909 |
| tk | Turkmen | 1 | 66,104 |
| bs | Bosnian | 1 | 63,018 |
| kk | Kazakh | 1 | 41,839 |
| sl | Slovenian | 1 | 22,844 |
| sk | Slovak | 1 | 12,997 |
| co | Corsican | 1 | 9,083 |
| gn | Guarani | 1 | 1,566 |
| bo | Tibetan | 1 | 579 |

Table 5: HALvest's splits.

| Domain | Code | # Documents | # mT5 Tokens |
|---|---|---|---|
| Humanities and Social Sciences | shs | 156,566 | 5,614,423,171 |
| Computer Science | info | 148,316 | 2,573,673,455 |
| Life Sciences | sdv | 115,744 | 3,145,323,780 |
| Engineering Sciences | spi | 102,751 | 2,254,653,825 |
| Physics | phys | 65,991 | 1,503,190,749 |
| Mathematics | math | 62,921 | 1,638,500,361 |
| Chemical Science | chim | 40,012 | 899,507,319 |
| Environmental Science | sde | 31,575 | 579,076,669 |
| Sciences of the Universe | sdu | 23,557 | 682,356,264 |
| Cognitive Science | scco | 11,772 | 227,487,096 |
| Statistics | stat | 10,579 | 184,678,350 |
| Quantitative Finance | qfin | 3,451 | 68,518,636 |
| Nonlinear Sciences | nlin | 1,972 | 30,694,088 |

Table 6: HALvest's domain statistics.

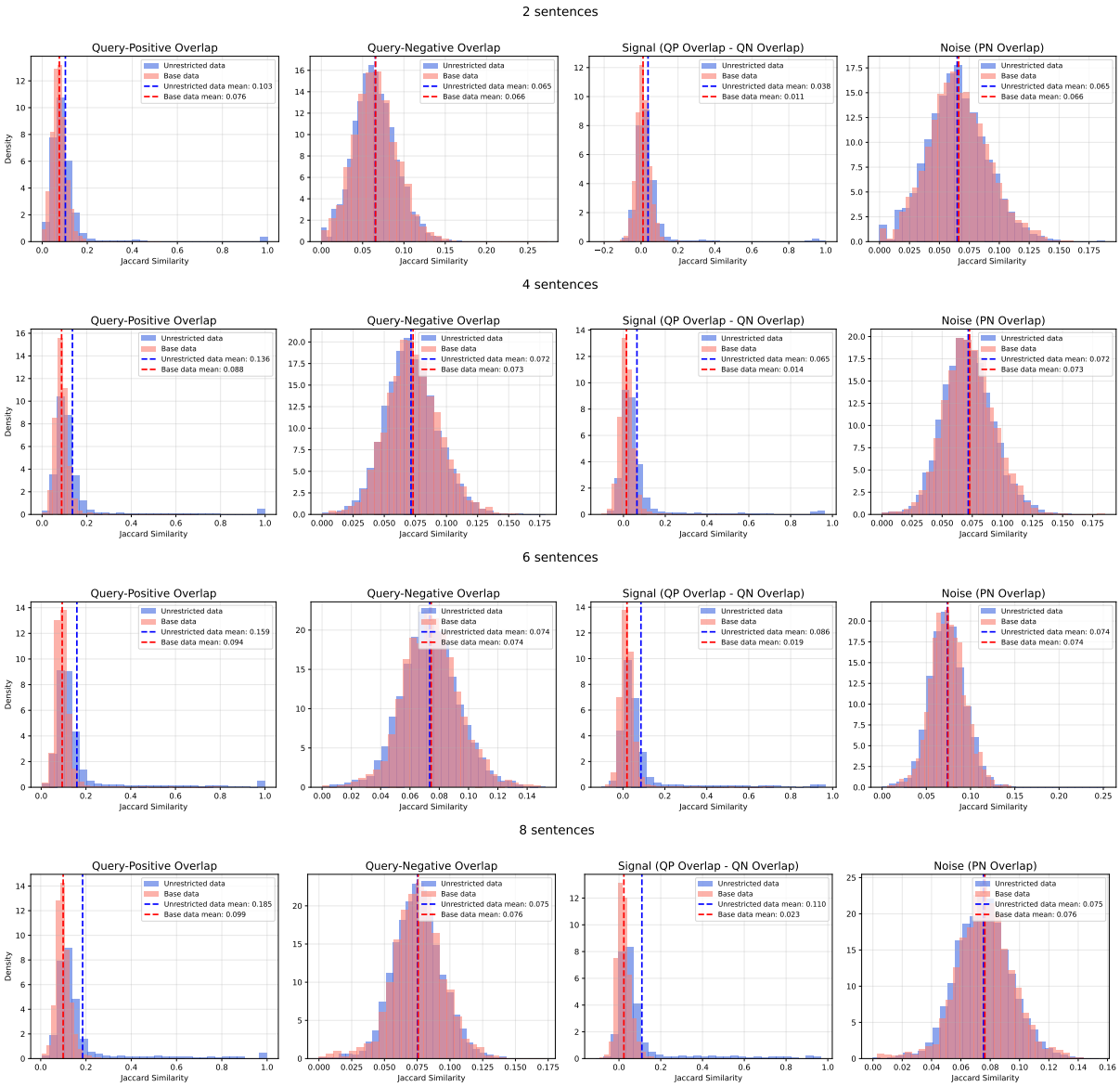


Figure 3: Signal and noise on every test splits before scaling. We split 10,000 examples by word and compute the Jaccard similarity between the query and the positive, as well as the query and the negative. We define the signal as the difference between the query/positive and the query/negative overlaps. The noise is the Jaccard similarity between the positive and the negative of a triplet.